Physics Ch1 Supplementary notes and exercise

1. Graph and proportion

Aim : To understand how to plot a graph properly. (labelling, correct scale, correct line, etc

To understand the information provided from the graph.

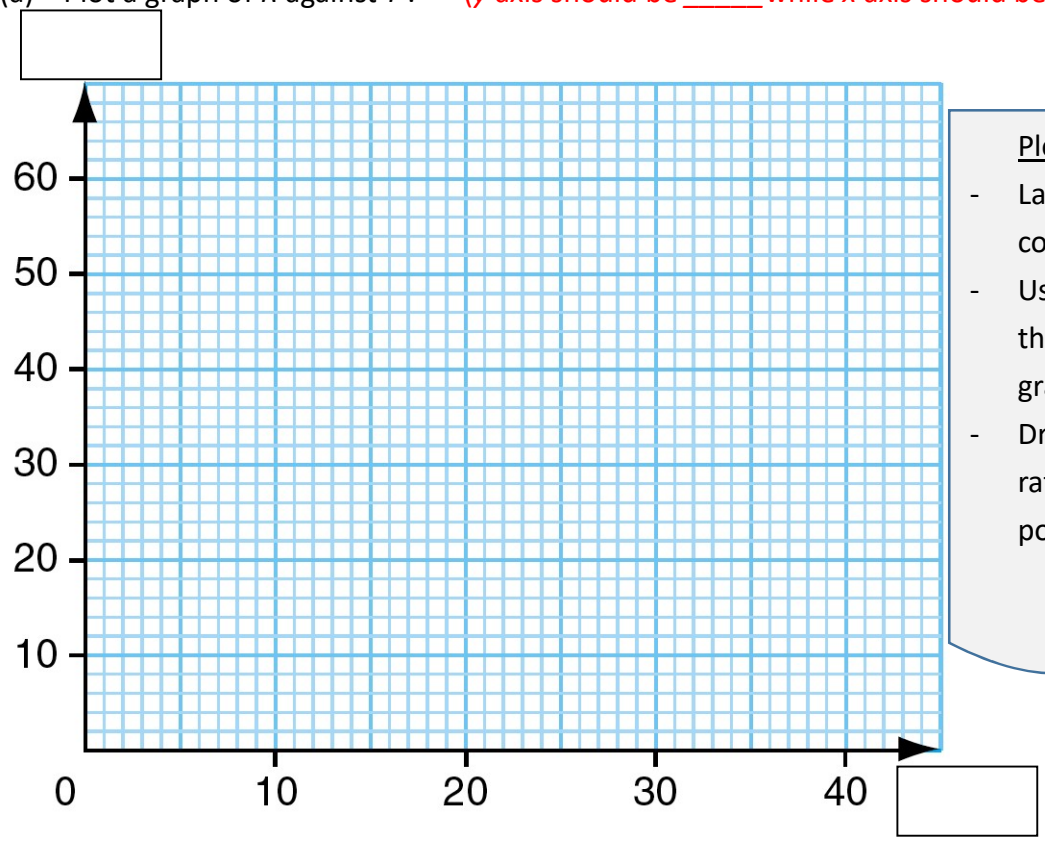
Students are able to use graphical method or proportion /ratio to solve problems of linear relationship.

Example

The resistance *R* of a resistance thermometer at different temperatures *T* is shown.

***T* / °C**	0	10	20	30	40
***R* / Ω**	12	23	35	50	61

(a) Plot a graph of *R* against *T* . (*y* axis should be _____ while *x* axis should be _____)



Please Check

- Label the axes with correct unit.
- Use crosses to mark the points on the graph
- Draw a smooth line rather than joining the points together.

The concept of slope will be introduced in Ch.9 in Book 3B

(b) What is the temperature when *R* = 40 Ω ?

Method 1 , From the graph, When *R* = 40 Ω, *T* = _____ °C

Method 2, (By proportion /ratio /slope)

$$\frac{Y_2 - Y_1}{X_2 - X_1} = \frac{Y_1 - Y_0}{X_1 - X_0} \text{ or } \frac{X_2 - X_1}{Y_2 - Y_1} = \frac{X_1 - X_0}{Y_1 - Y_0}$$ (concept of ratio of similar triangle or concept of slope)

Take two points (62, 40) and (40, 15) as reference points. And then find out the coordinate of the point (40, T).

By $$\frac{X_2 - X_1}{Y_2 - Y_1} = \frac{X_1 - X_0}{Y_1 - Y_0} \Rightarrow \frac{R_{62} - R_{30}}{R_{40} - R_{30}} = \frac{T_{40} - T_{15}}{T - T_{15}}$$

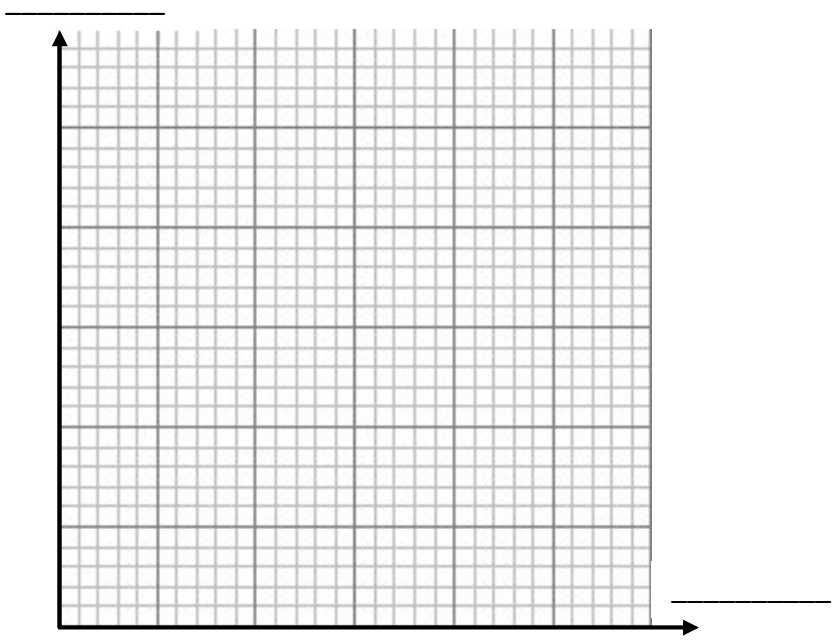
Then $$\frac{T - 15}{40 - 30} = \frac{40 - 15}{62 - 30} \Rightarrow T = 22.8^{o}C$$

Exercise

1. The liquid column in an unmarked liquid in glass thermometer has different lengths L when the temperature bulb is put into different beakers of water with different temperature *T* as below.

***T* / °C**	0	20	40	60	80
L* / *cm	5	9.5	13	16.5	21

(a) Plot a graph of *L* against *T* .



(b) Find the length of the liquid column L at 50 °C,

(i) by using the graph ;

(ii) by using calculation method.

(c) Find the temperature when the liquid column L is 30 cm.

(Hint : use $\frac{Y_2 - Y_0}{Y_1 - Y_0} = \frac{X_2 - X_0}{X_1 - X_0}$)

2. The length of the mercury thread in a mercury-in-glass thermometer are 2 cm and 8 cm at 0 °C and 100 °C respectively.

   (a) Find the temperature when the length of the mercury thread is 6 cm.

   (b) Find the length of the mercury thread when the temperature is 45 °C.

3. The markings on a calibrated thermometer have faded away. We know that the column lengths for 0 °C and 50 °C are 4 cm and 10 cm respectively. What is the column length for 70 °C?

4. A certain wire has a resistance of 9Ω at 0 °C and 14Ω at 150 °C. When the wire is placed in liquid oxygen, the resistance falls to 3Ω. Estimate the temperature in °C of the liquid oxygen.

2. Scientific notation and unit conversion

Power equation $P = \frac{Q}{t}$ i.e. $1\text{W} = \frac{1J}{1s}$

Make Energy (Q) as subject, $Q = P \times t$ i.e. $1\text{J} = 1\text{W} \times 1\text{s}$ or 1Ws

Energy can be express as kWh.

$\because Q = P \times t$

$\therefore 1kWh = 1\,kW \times 1\,hr = (1000\,J/s) \times (3600\,s) = 3600000\,J = 3.6 \times 10^6\,J$

Exercise

1. Express the following terms in Joule J with correct notation.
   (a) 15 kwh

   $15\,kWh = 15 \times 1\,kWh$

   $= 15 \times$ _____J

   $=$ _____ $\times 10$——J

   (b) 2.5 kWh

   $2.5\,kWh = 2.5 \times 1\,kWh$

   $=$

   (c) 7 kwh

2. Express the following terms in kWh
   (a) 7200000 J

   $\because 3600000J = 1kWh$

   $\therefore \frac{7200000\,J}{3600000\,J} = \frac{(X)kWh}{1\,kWh}$

   X = _____

   $\therefore$ 7200000 J = _____ kWh

   (b) 18 000 000 J

   $\therefore \frac{18\,000\,000\,J}{3\,600\,000\,J} = \frac{(X)kWh}{1\,kWh}$

   X = _____

   $\therefore$ 18 000 000 J = _____ kWh

   (c) $16.2 \times 10^7 J$